

GrounUp Enterprise Specification

Phase 02 - Core Platform

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Purpose

Reusable secure runtime foundation for all GrounUp capabilities.

Platform Domains

Identity & Access: Authentication, authorization, sessions, service identities, and device trust

Tenancy: Tenant context, isolation, provisioning, and regional placement

Platform Services: Files, search, notifications, scheduler, jobs, cache, metadata, tags, and audit

Workflow & Rules: Workflow orchestration, rules execution, approvals, state transitions, and timers

Integration: API gateway, webhooks, connectors, imports, exports, and event contracts

Mobile & Offline: Mobile shell, offline data packs, sync, conflict resolution, and field resilience

Observability: Logs, metrics, traces, health checks, alerting, and operational dashboards

Security: Encryption, secrets, key management, secure configuration, threat controls, and recovery

Localization: Time zones, languages, currencies, units, regional formats, and calendars

AI Platform: Agent registry, model gateway, tool execution, evidence, policy, and AI event logging

Functional Requirements

CPF-000001 - Authentication framework. The platform shall support pluggable authentication providers including passwordless, SSO, MFA, and enterprise identity federation. *Acceptance:* A tenant can configure approved providers and authentication policies without code changes.

CPF-000002 - Session management. The platform shall issue, refresh, revoke, and expire sessions with configurable risk and inactivity policies. *Acceptance:* Revoked or expired sessions cannot access APIs or user interfaces.

CPF-000003 - Service identities. The platform shall support non-human identities for integrations, jobs, and AI agents. *Acceptance:* Every service identity has explicit owner, scope, credentials, rotation, and audit history.

CPF-000004 - Authorization service. All protected platform actions shall use a centralized authorization decision service. *Acceptance:* Every decision returns allow or deny, matched policy, effective scope, and correlation ID.

CPF-000005 - Device trust. The platform shall register device posture and support risk-based access controls. *Acceptance:* Policy can require trusted devices for sensitive actions.

CPF-000006 - Tenant context propagation. Tenant context shall propagate through API, event, job, cache, search, file, and AI operations. *Acceptance:* No platform service may process a governed request without verified tenant context.

CPF-000007 - Tenant data isolation. All storage and query paths shall enforce tenant isolation. *Acceptance:* Cross-tenant access attempts are blocked, logged, and testable.

CPF-000008 - Regional data placement. Tenants shall support region selection and data-residency constraints. *Acceptance:* Configured regional placement is visible and enforced for governed data classes.

CPF-000009 - Provisioning automation. Tenant provisioning shall create required namespaces, defaults, indexes, storage, keys, and baseline configurations. *Acceptance:* Provisioning produces a validation report and rollback plan.

CPF-000010 - Global identifier service. The platform shall provide globally unique, non-guessable, immutable identifiers for governed records. *Acceptance:* Identifiers remain stable across versions, imports, exports, and integrations.

CPF-000011 - Metadata framework. Entities and fields shall support extensible metadata definitions without ungoverned schema changes. *Acceptance:* Metadata is versioned, validated, permissioned, and queryable.

CPF-000012 - Universal tagging. Users and automation shall apply governed tags to supported objects. *Acceptance:* Tags support ownership, scope, synonyms, lifecycle, and search.

CPF-000013 - File and document service. The platform shall store, version, classify, scan, preview, and secure files and documents. *Acceptance:* Every file version has checksum, owner, classification, retention, and access policy.

CPF-000014 - Search service. The platform shall provide permission-aware full-text and structured search. *Acceptance:* Search results never expose unauthorized records or snippets.

CPF-000015 - Notification service. The platform shall send in-app, email, SMS, push, and webhook notifications through governed templates and preferences. *Acceptance:* Every notification has delivery status, retry, deduplication, and audit.

CPF-000016 - Scheduler service. The platform shall schedule one-time and recurring tasks with tenant-aware execution. *Acceptance:* Schedules are versioned, observable, cancellable, and protected from duplicate execution.

CPF-000017 - Background job service. Long-running work shall execute through durable background jobs. *Acceptance:* Jobs support priority, retry, idempotency, progress, cancellation, dead-letter handling, and audit.

CPF-000018 - Distributed cache. The platform shall provide a tenant-aware cache with explicit key ownership, TTL, invalidation, and fallback behavior. *Acceptance:* Cache failure cannot corrupt source-of-truth data.

CPF-000019 - Configuration service. Runtime configuration shall be centrally resolved, versioned, validated, and auditable. *Acceptance:* Services can read effective configuration and subscribe to approved changes.

CPF-000020 - Feature flag service. Capabilities shall support tenant, user, percentage, environment, and effective-date feature flags. *Acceptance:* Feature rollout and rollback are controlled without redeployment.

CPF-000021 - Workflow engine. The platform shall orchestrate durable stateful workflows with timers, approvals, retries, and compensation. *Acceptance:* Workflow state survives service restart and can be inspected.

CPF-000022 - Rules engine. The platform shall evaluate versioned deterministic rules against declared inputs. *Acceptance:* Rule results are reproducible, explainable, and attributable to a rule version.

CPF-000023 - State machine framework. Governed entities shall support explicit state machines and allowed transitions. *Acceptance:* Invalid transitions are blocked and valid transitions emit events.

CPF-000024 - Timer and SLA engine. Workflows shall support due dates, timers, escalation, pause, resume, and calendar-aware SLAs. *Acceptance:* Timer behavior remains correct across time zones and downtime.

CPF-000025 - API gateway. External and internal APIs shall pass through a governed gateway. *Acceptance:* The gateway enforces authentication, authorization, versioning, throttling, request limits, and correlation.

CPF-000026 - API versioning. APIs shall use explicit versioning and deprecation policies. *Acceptance:* Breaking changes require a new supported version and migration notice.

CPF-000027 - Webhook service. The platform shall publish signed, retryable, tenant-scoped webhooks. *Acceptance:* Consumers can verify signature, replay protection, event ID, and delivery attempt.

CPF-000028 - Event bus. Platform domains shall publish durable versioned events through an event bus. *Acceptance:* Events have schema, tenant, producer, timestamp, correlation, causation, and idempotency keys.

CPF-000029 - Connector framework. The platform shall support governed connectors with credentials, scopes, mappings, health, and sync state. *Acceptance:* Connector actions are observable and independently disableable.

CPF-000030 - Import engine. The platform shall import structured data using mappings, validation, previews, errors, and resumable batches. *Acceptance:* No invalid row is silently accepted.

CPF-000031 - Export engine. The platform shall export authorized data to governed formats with filters and classification controls. *Acceptance:* Exports record requester, criteria, file, checksum, retention, and expiration.

CPF-000032 - Mobile application shell. The platform shall provide a reusable mobile shell for field workflows. *Acceptance:* The shell supports secure login, navigation, camera, location, notifications, and local storage.

CPF-000033 - Offline data packs. Authorized users shall download scoped offline data packs for field work. *Acceptance:* Offline packs are encrypted, time-limited, versioned, and revocable.

CPF-000034 - Offline transaction queue. Offline actions shall be queued locally with durable ordering and status. *Acceptance:* Queued changes preserve original user, device, timestamp, and correlation.

CPF-000035 - Synchronization engine. The platform shall synchronize offline and online changes with deterministic conflict handling. *Acceptance:* Sync results identify applied, rejected, merged, and manual-review changes.

CPF-000036 - Attachment synchronization. Files captured offline shall upload resumably with checksums and classification. *Acceptance:* Duplicate or corrupt uploads are detected.

CPF-000037 - Structured logging. All services shall emit structured logs with tenant-safe correlation fields. *Acceptance:* Logs exclude prohibited secrets and sensitive payloads by default.

CPF-000038 - Metrics and health. Services shall expose health, capacity, latency, error, and dependency metrics. *Acceptance:* Health checks distinguish startup, readiness, liveness, and degraded state.

CPF-000039 - Distributed tracing. Requests, events, jobs, workflows, and AI actions shall support distributed tracing. *Acceptance:* A correlation can be followed across service boundaries.

CPF-000040 - Alerting framework. Operational thresholds shall create deduplicated alerts with ownership and escalation. *Acceptance:* Alerts contain runbook, impact, severity, and supporting evidence.

CPF-000041 - Encryption in transit. All supported network communication shall use approved encrypted transport. *Acceptance:* Insecure transport is rejected except isolated approved development environments.

CPF-000042 - Encryption at rest. Governed data shall be encrypted at rest with managed keys. *Acceptance:* Key use, rotation, access, and failure are auditable.

CPF-000043 - Secrets management. Credentials and secrets shall be stored only in approved secret stores. *Acceptance:* Secrets are never persisted in source code, ordinary logs, or user-visible configuration.

CPF-000044 - Malware and content scanning. Uploaded content shall be scanned and quarantined based on policy. *Acceptance:* Unsafe content cannot be opened or distributed before disposition.

CPF-000045 - Backup and restore. The platform shall perform encrypted backups and tested restoration. *Acceptance:* Restore tests prove stated recovery point and recovery time objectives.

CPF-000046 - Disaster recovery. Critical services shall have documented failover, recovery, and communication procedures. *Acceptance:* Recovery exercises produce evidence and corrective actions.

CPF-000047 - Rate limiting and abuse control. The platform shall rate-limit and detect abusive API, login, export, and automation behavior. *Acceptance:* Controls can be scoped and do not expose cross-tenant information.

CPF-000048 - Time zone framework. All timestamps shall be stored in UTC and rendered using declared user or project time zones. *Acceptance:* Scheduling and due dates remain correct through daylight-saving transitions.

CPF-000049 - Unit framework. The platform shall support US customary and metric units with governed conversion and precision. *Acceptance:* Every quantity stores value, unit, precision, and conversion provenance.

CPF-000050 - Currency framework. Monetary values shall store currency and support governed conversion rates where required. *Acceptance:* No arithmetic combines different currencies without an explicit conversion.

CPF-000051 - Language and locale. The platform shall support locale-aware text, numbers, dates, and translations. *Acceptance:* Users can select supported locale without changing stored canonical values.

CPF-000052 - Model gateway. AI model calls shall pass through a governed model gateway. *Acceptance:* The gateway enforces model policy, tenant context, limits, logging, redaction, and fallback.

CPF-000053 - Agent registry. AI agents shall be registered with owner, purpose, version, permissions, tools, and evaluation status. *Acceptance:* Unregistered or disabled agents cannot execute.

CPF-000054 - Tool execution service. AI tools shall execute through permissioned, validated, observable adapters. *Acceptance:* Every tool invocation records input summary, output summary, result, duration, and correlation.

CPF-000055 - Prompt and policy registry. Prompts and AI policies shall be versioned, approved, testable, and rollbackable. *Acceptance:* Production execution identifies exact prompt and policy versions.

CPF-000056 - Evidence and citation service. AI outputs shall attach source references and evidence where required. *Acceptance:* Evidence remains resolvable to the authorized source version.

CPF-000057 - AI evaluation framework. Agents and prompts shall be evaluated for accuracy, safety, authorization, consistency, and business acceptance. *Acceptance:* Failed release gates prevent promotion.

CPF-000058 - AI usage and cost metering. The platform shall meter AI requests, tokens, tools, latency, errors, and cost by tenant and capability. *Acceptance:* Usage reports reconcile to gateway records.

CPF-000059 - Platform event audit. Material platform events shall be preserved in an immutable audit trail. *Acceptance:* Events can be correlated to request, actor, service, object, and tenant.

CPF-000060 - Data portability. Authorized tenants shall export core data and attachments in documented formats. *Acceptance:* Exports are complete, traceable, and do not include unauthorized shared data.

Canonical Entities

ID	Entity	Purpose	Lifecycle	Owner
ENT-P02-001	PlatformRequest	Normalized inbound request context	received; processing; completed; failed	Platform Architecture
ENT-P02-002	Session	Authenticated interactive or service session	active; challenged; revoked; expired	Identity Platform
ENT-P02-003	Device	Registered client device	new; trusted; restricted; revoked	Mobile Security
ENT-P02-004	AuthorizationDecision	Explainable access decision	recorded	Security Platform
ENT-P02-005	ServiceIdentity	Non-human platform identity	draft; active; suspended; retired	Identity Platform
ENT-P02-006	MetadataDefinition	Governed extension field definition	draft; active; deprecated	Data Architecture
ENT-P02-007	MetadataValue	Value attached to governed object	active; superseded	Data Architecture
ENT-P02-008	TagDefinition	Governed tag definition	draft; active; retired	Information Architecture
ENT-P02-009	TagAssignment	Tag applied to an object	active; removed	Information Architecture
ENT-P02-010	FileObject	Logical file identity	active; archived; deleted	Records Platform
ENT-P02-011	FileVersion	Immutable content version	uploading; scanning; accepted; quarantined; rejected	Records Platform
ENT-P02-012	SearchDocument	Permission-aware indexed representation	queued; indexed; stale; deleted	Search Platform
ENT-P02-013	Notification	Outbound communication instance	queued; sent; delivered; failed; cancelled	Notification Platform
ENT-P02-014	NotificationTemplate	Versioned message template	draft; active; retired	Notification Platform
ENT-P02-015	Schedule	One-time or recurring execution plan	draft; active; paused; completed; cancelled	Scheduler Platform
ENT-P02-016	BackgroundJob	Durable asynchronous work item	queued; running; retry; succeeded; failed; cancelled; dead-lettered	Job Platform
ENT-P02-017	WorkflowInstance	Durable workflow execution	running; waiting; completed; failed; cancelled	Workflow Platform
ENT-P02-018	WorkflowDefinition	Versioned workflow graph	draft; approved; active; retired	Workflow Platform
ENT-P02-019	RuleDefinition	Versioned deterministic rule	draft; approved; active; retired	Rules Platform
ENT-P02-020	RuleExecution	Recorded rule evaluation	recorded	Rules Platform
ENT-P02-021	DomainEvent	Versioned event envelope	published; archived	Integration Platform
ENT-P02-022	EventDelivery	Consumer delivery state	pending; delivered; retry; dead-lettered	Integration Platform
ENT-P02-023	WebhookSubscription	Tenant outbound webhook configuration	draft; active; paused; disabled	Integration Platform
ENT-P02-024	WebhookDelivery	Signed delivery attempt	queued; delivered; retry; failed; dead-lettered	Integration Platform
ENT-P02-025	ConnectorInstance	Configured external system connector	draft; active; paused; error; retired	Integration Platform
ENT-P02-026	ImportBatch	Governed import execution	uploaded; validating; preview; imported; rejected; failed	Data Exchange
ENT-P02-027	ExportPackage	Governed export artifact	requested; generating; available; exported; failed	Data Exchange
ENT-P02-028	OfflinePack	Encrypted field data package	building; available; downloaded; expired; failed	Mobile Platform
ENT-P02-029	SyncTransaction	Offline-originating change	queued; received; applied; rejected; cancelled	Mobile Platform
ENT-P02-030	ConflictCase	Manual or governed sync conflict	open; resolved; rejected	Mobile Platform
ENT-P02-031	FeatureFlag	Versioned rollout control	draft; active; paused; retired	Release Management
ENT-P02-032	SecretReference	Reference to protected secret	active; rotation_due; revoked	Security Engineering
ENT-P02-033	BackupSet	Encrypted recovery point	creating; available; expired; failed	Resilience Engineering
ENT-P02-034	AIAgentDefinition	Registered AI agent version	draft; review; approved; active; suspended	AI Platform
ENT-P02-035	AIExecution	Governed AI request execution	queued; running; completed; failed; blocked	AI Platform
ENT-P02-036	AIToolInvocation	Permissioned AI tool call	requested; authorized; executed; denied; failed	AI Platform

Core Workflows and Engines

WF-P02-001 - Authenticated Request. Validate transport; authenticate; resolve tenant; authorize; execute; audit; respond

WF-P02-002 - File Upload and Acceptance. Create upload session; stream chunks; verify checksum; scan; classify; create immutable version; index

WF-P02-003 - Search Indexing. Read source; create permission tokens; transform fields; index; mark version

WF-P02-004 - Background Job Execution. Validate payload; claim lease; execute; checkpoint; retry; dead-letter; notify

WF-P02-005 - Durable Workflow Execution. Load definition; execute steps; persist state; wait timers/events; retry or compensate

WF-P02-006 - Webhook Delivery. Build payload; sign; deliver; capture response; backoff; dead-letter; notify

WF-P02-007 - Import Batch. Parse; map; validate rows; preview; approve; write idempotently; report errors

WF-P02-008 - Offline Synchronization. Authenticate device; upload queue; validate base versions; apply or resolve; download deltas; confirm

WF-P02-009 - Feature Rollout. Define targeting; approve; enable small cohort; monitor; expand or rollback

WF-P02-010 - Secret Rotation. Create new secret; update references; verify; revoke old; audit

WF-P02-011 - Backup and Restore Verification. Create encrypted backup; verify checksum; restore isolated; run validation; record evidence

WF-P02-012 - AI Execution. Resolve agent/version; authorize; redact; call model; execute tools; attach evidence; meter; audit

ENG-P02-001 - Authorization Decision Engine. Deterministic and side-effect free

ENG-P02-002 - Configuration Resolution Engine. Highest authorized active override wins

ENG-P02-003 - Workflow Orchestration Engine. Persist before external side effects

ENG-P02-004 - Rules Evaluation Engine. Same inputs and version yield same result

ENG-P02-005 - Job Dispatch Engine. At-least-once processing

ENG-P02-006 - Search Indexing Engine. Authorization filtered at query time

ENG-P02-007 - Notification Routing Engine. Honor mandatory notifications over opt-out

ENG-P02-008 - Synchronization Engine. Deterministic conflict resolution

ENG-P02-009 - Unit Conversion Engine. Dimension compatibility required

ENG-P02-010 - Currency Conversion Engine. No implicit conversion

ENG-P02-011 - AI Policy Engine. Intersection of user, agent, tenant, and tool policy

ENG-P02-012 - AI Cost Metering Engine. Reconcile to gateway execution

AI Platform

Production AI uses the model gateway, registered agents, permissioned tools, evidence, evaluation, cost metering, and complete audit.